

Real-time Data Documentation

Overview

The documentation pertains to the real-time data are contained in MASTERFILE_Real_Time_Data.xlsx. This data package is updated monthly using real time data available on the last day of each month at 11:59 pm EST.

Please cite the appropriate following sources when using this data:

- The real-time data for the global crude oil market is from Benyo et al. (2026). Vintages prior to 2010M11 come from Baumeister and Kilian (2012).
- The real-time data for the Refiner's Acquisition Cost of Crude Oil was used in Ellwanger and Snudden (2023a) and Ellwanger and Snudden (2023b).

This documentation contains two main sections: Section I pertains to the Masterfile of real-time variables; Section II pertains to the treatment of the real time data.

Note that monthly average and end-of-month spot prices for Brent and WTI crude oil are not subject to real-time revision and are reported in a separate data package.

Section I – Masterfile of Real-time Variables Spreadsheet Documentation

The spreadsheet contains 5 real-time variables, and all the variables are at a monthly frequency: (i) Nominal Refiner's Acquisition Cost of Crude Oil, imported (RAC), (ii) World Oil Production, (iii) US Crude Oil Inventories, (iv) US Petrol Inventories, (v) and OECD Petroleum Inventories Real Time.

Descriptions of Data Series

Nominal RAC

This is the Refiner's Acquisition Cost of Crude Oil expressed in dollars per barrel. The typical delay of this data series is 2 months, and this data series is usually released on the last week of the month. The data is retrieved from U.S. Energy Information Administration's (EIA) Monthly Energy Review (MER). Data for 1973 follows Mork (1989). Data vintages prior to 2010M11 come from Baumeister and Kilian (2012) and do not reflect all real-time revisions. Historical vintages thereafter contain all revisions using both regular collection and the internet archive service provided by <https://archive.org/web/>. Imp

US Crude Oil/Petrol Inventories

The Total Crude Oil/Petrol Stocks of the United States is expressed in million barrels. The data is retrieved from U.S. Energy Information Administration's (EIA) Monthly Energy Review (MER). The

typical delay of this data series is 1 month, and this data series is usually released on the last week of the month. Data prior to 2010M11 comes from Baumeister and Kilian (2012). Historical vintages thereafter were updated monthly using both regular collection and the internet archive service provided by <https://archive.org/web/>. For the months the full historical vintages are not available, the pdf summary is used to fill in missing variables.

OECD Petroleum Inventories

This is the OECD petroleum stocks include crude oil and lease condensates (with strategic reserves), natural gas plant liquids, refinery feedstocks, additives and oxygenates, other hydrocarbons, and refined petroleum products of OECD, expressed in million barrels. Data prior to 2010M11 comes from Baumeister and Kilian (2012). Historical vintages thereafter were updated monthly using both regular collection and the internet archive service provided by <https://archive.org/web/>. After 2019M06 the MER stopped reporting this data and it began to be retrieved from the EIA website under the [international data browser](#). The typical delay of this data is 3 months, and this data series is usually released on the first week of the month.

World Oil Production

This is the monthly data for the world's crude oil including lease condensate production expressed in millions of barrels per day. The data is retrieved from U.S. Energy Information Administration's (EIA) Monthly Energy Review (MER). The data is updated on the first week of every month and the data is delayed by 3 months. Data prior to 2010M11 comes from Baumeister and Kilian (2012). Historical vintages thereafter were updated monthly using both regular collection and the internet archive service provided by <https://archive.org/web/>.

The MER stopped reporting this series in the MER on 2019M06. The data series was not able to be retrieved again until 2021M01 when it became available on the EIA website under the international data browser (<https://www.eia.gov/international/data/world>). The data from the browser has a delay of an additional month, making the data delay now four months. The four-month delay is assumed to begin for the 2019M07 vintage. Moreover, during this blackout period the series experienced a revision going back to 2005M02. Following the real-time method, missing vintages between 2019M07 to 2021M12 use information available on 2019M06 and missing observations are imputed using the growth rate of the data in the 2021M01 vintage.

US CPI

The real-time data for the monthly seasonally adjusted U.S. consumer price index for all urban consumers is available from the Economic Indicators published by the Council of Economic Advisers from the FRASER database of the Federal Reserve Bank of St. Louis and from the macroeconomic real-time database of the [Federal Reserve Bank of Philadelphia \(PCPI\)](#). The typical delay of this data series is 1 month, and this data series is usually released in the middle of the month.

Section II – Completeness of the Realtime Vintages

We distinguish the data by 5 different colors: Green, Grey, Blue, Orange, Purple, based on the way that the data is collected for a particular vintage. A green vintage implies that the full historical vintage for that point in time was observed. For example, all historical observations were downloaded from the MER using the spreadsheet provided. Data highlighted in gray implies the excel files of the full vintage were not available, but that sufficient information was provided in the pdf of the MER to determine the exact values at those points in time. Blue implies that a date range had some revisions, and it is possible to deduce the revisions using surrounding vintages. The orange shaded series is the case where revisions or real time data could not be observed and used information from previous vintages to impute the missing values. In these cases, the vintages rely on the information that would be available at that point in time. Finally, purple values are used to show cases in Baumeister and Kilian (2012) where historical revisions were known to occur but were not reflected in the data.

Color Coding Legend

Color	Description
Green	The full historical vintage is directly observed for that particular month. This implies that the entire historical series was downloaded at that point in time.
Grey	The data was constructed from the pdf files of the data rather than the original spreadsheet. These cases include all revisions.
Blue	This indicates that the data range had some revisions made but we were able to deduce the data exactly using surrounding vintages.
Orange	This is the case where we were not able to observe the actual data itself and had to use information to that point in time to fill in missing values.
Purple	Used to show in Baumeister and Kilian (2012) where revisions occurred but were not reflected in the data.

Table 1. Real-time Variables (Filename: MASTERFILE_Real_Time_Data)						
<u>Series</u>	<u>Unit of Measurement</u>	<u>Start-Date</u>	<u>Vintage</u>	<u>Data Delay (months)</u>	<u>Release Time (of the month)</u>	<u>Data Sources</u>
Nominal RAC	Dollars per Barrel	1973 - 01	1991M01 –	1991M01 – 2005M07: 3 2005M08 – : 2 2006M09: 3	N/A	Imputed, Mork (1989, p. 741)
		1974 - 01	1991M01 –		Last week of month	Crude Oil Price Summary, Monthly Energy Review from the EIA
World Oil Production	Thousand Barrels Per Day	1973 - 01	1991M01 – 2019M06	3	Last week of month	Crude Oil Price Summary, Monthly Energy Review from the EIA
		1973 - 01	2019M07 –	4	First week of month	<i>International Petroleum and Other Liquids</i> from the EIA International Data Browser
US Crude Oil/Petrol Inventories	Million Barrels	1973 - 01	1991M01 –	1	Last week of month	Petroleum Stocks, Monthly Energy Review from the EIA
OECD Petroleum Inventories	Million Barrels	1988 - 01	1991M01 – 2019M06	1991M01 – 1991M10: 6 1991M11 – 2002M10: 5 2002M11 – 2016M06:4 2016M07 – : 3	Last week of month	Crude Oil Price Summary, Monthly Energy Review from the EIA
		1980 - 01	2019M07 –		First week of month	<i>Monthly OECD petroleum and other liquids stocks</i> from the EIA International Data Browser
Consumer Price Index (CPI)	All Urban Consumers, seasonally adjusted	1973 – 01	1991M01 –	1	Middle of the month	<i>Macroeconomic real-time database of the Federal Reserve Bank of Philadelphia (PCPI).</i>

References

- Baumeister, C., and Kilian, L. (2012). Real-time forecasts of the real price of oil. *Journal of Business & Economic Statistics*, 30(2): 326–336.
- Benyo, E., R. Ellwanger, and S. Snudden (2026) A reappraisal of real-time forecasts of the real price of oil.” *Economic Inquiry*, 64(1): 167–176.
- Ellwanger, R. and S. Snudden, (2023a). Forecasts of the Real Price of Oil Revisited: Do they Beat the Random Walk? *Journal of Banking and Finance*, 154(106962): 1–8.
- Ellwanger, R. and Snudden, S. (2023b). Futures Prices are Useful Predictors of the Spot Price of Crude Oil. *The Energy Journal*, 44(4): 45–62.
- Mork, K. A. (1989). Oil and the macroeconomy when prices go up and down: an extension of Hamilton's results. *Journal of Political Economy*, 97(3): 740–744.